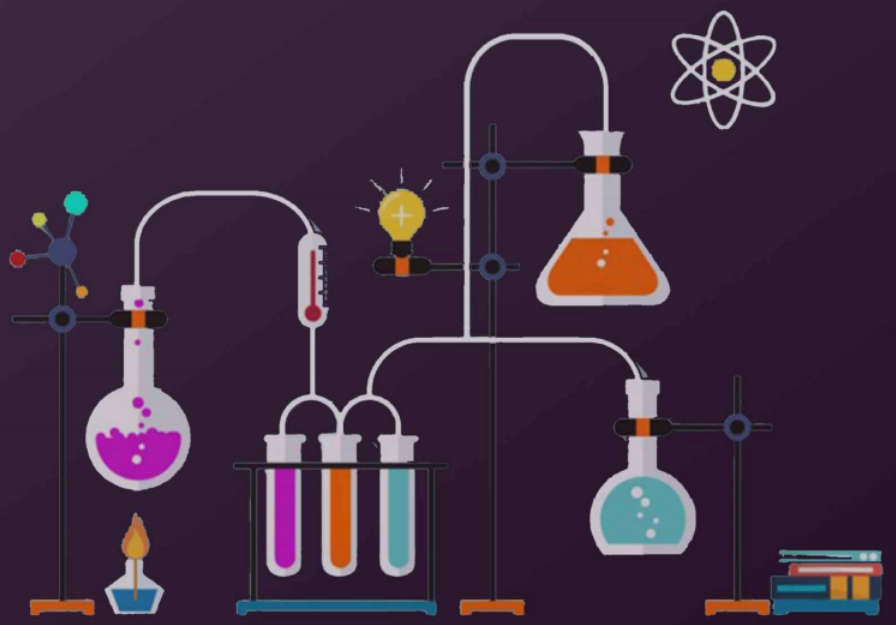
CHEMISTRY

Redox Reaction



SHEET

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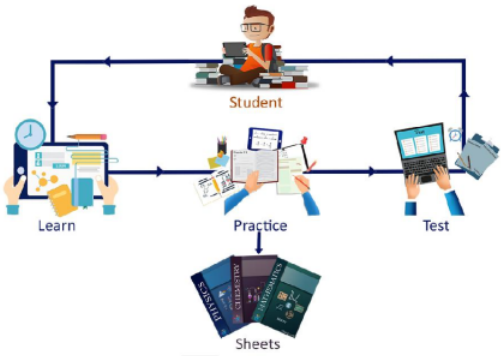
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Redox Reactions	1			
2	Oxidation and Reduction	2 - 17	1-4, 10-18, 20,21,27,28	1-20	1,2,16,17, 19
3	Types of Redox Reactions	18 - 21	5,6,22,23		3
4	Balancing of Redox reaction	22 - 24	24	21-27	12,13, 20
5	Ion-electron Method of balancing redox reactions	25 - 26	25,29	28,29	21,22
6	Equivalent Weight	27 - 36	26	30-39	
7	Normality And It's Applications	37 - 41	7,19		7,8,18
8	Titration, Iodometry, Iodimetry	42 - 45	8,9	40,41	5,9-11,15
9	ACC	46 - 53		42-45	4,6,14

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Redox Reaction

Q1. Redox Means-
(A) Oxidation
(B) Reduction
(C) Oxidation + Reduction
(D) None of These

Oxidation and Reduction

Q2. If H_2O_2 the oxidation state of oxygen is ?
(A) –2 (B) +1 (C) –1 (D) –1/2

Q3. Which of the following is a reducing agent in the given reaction :-

$HgCl_2 + Sn \longrightarrow SnCl_2 + Hg$

(A) Sn (B) Hg
(C) Cl (D) Both (A) & (B)

Q4. Oxidation number of Cl in $CaOCl_2$ (bleaching powder is)
(A) Zero, since it contains Cl_2
(B) –1, since it contains Cl^-
(C) +1, since it contains ClO^-
(D) +1 and –1 since it contains ClO^- and Cl^-

Q5. Which of the following groups of molecules act both as oxidising agent as well as reducing agent:-
(A) $KMnO_4$, O_3, SO_3
(B) $HClO_4$,HNO_2,H_2O_2
(C) HNO_2, SO_2, H_2O_2
(D) HNO_3, SO_2, H_2SO_4

Q6. Which of following statement are correct :-
(A) Oxidation number of Os in OsO_4 is +8
(B) Oxidation number of S in H_2SO_5 is +6
(C) Oxidation number of sulphur in $H_2S_2O_8$ is +6
(D) All of these

Q7. The oxidation states of S-atoms in Caro's and Marshell's acids are :-
(A) +6, +6 (B) +6, +4
(C) +6, –6 (D) +4, +6

Q8. The oxidation no. of oxygen atom in O_2^{2-} ion is :-
(A) –1 (B) –2
(C) –3 (D) –5

Q9. Which of these substances is a good reducing agent:-
(A) NaOCl (B) HI
(C) $FeCl_3$ (D) KBr

Q10. The oxidation state of S in $Na_2S_4O_6$ is :
(A) +2.5
(B) +2 and +3 (two S have +2 and other two have +3)
(C) +2 and +3 (three S have + 2 and one S has +3)
(D) +5 and 0 (two S have +5 and the other two S have 0)

Q11. Oxidation number of oxygen in F_2O is :
(A) +1 (B) +2
(C) –1 (D) –2

Q12. An element, which never has a positive oxidation state in any of its compounds, is:
(A) Boron (B) Oxygen
(C) Chlorine (D) Fluorine

Q13. The compound which could not act both as oxidising as well as reducing agent is:-

(A) SO_2 (B) MnO_2

(C) Al_2O_3 (D) HNO_2

Q14. In the reaction,

$Cl_2 + OH^- \rightarrow Cl^- + ClO_4^- + H_2O$,

chlorine is :

(A) Oxidised

(B) Reduced

(C) Oxidised as well as reduced

(D) Neither oxidised nor reduced

Q15. When P reacts with caustic soda, the products are PH_3 and NaH_2PO_2. The reaction is an example of :

(A) Oxidation

(B) Reduction

(C) Both oxidation and reduction

(D) Neutralisation

Q16. The oxidation number of Pt in $[Pt(C_2H_4)Cl_3]^-$ is-

(A) +1 (B) +2 (C) +3 (D) + 4

Q17. Consider the following oxyacids of phosphorus :

(A) Orthophosphoric acid

(B) Hypophosphoric acid

(C) Orthophosphorous acid

(D) phosphine

In which of these acids, P is not present in + 4 oxidation state ?

(A) 2 and 3 (B) 2, 3 and 4

(C) 3 and 4 (D) 1, 3 and 4

Types of Redox Reactions

Q18. Which of the following is not a redox reaction?

(A) $Cu + S \rightarrow CuS$

(B) $Cu^{2+} + S^{2-} \rightarrow CuS$

(C) $Zn + CuSO_4 \rightarrow + ZnSO_4 + Cu$

(D) All of these

Q19. Which of the following process represents disproportionation ?

(A) $Cu + 4HNO_3 \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$

(B) $3I_2 + 6OH^- \rightarrow IO_3^- + 5I^- + 3H_2O$

(C) $Cl_2 + I_2 \rightarrow 2ICl$

(D) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Q20. The metals which can undergo disproportionation are -

a. Sn b. Na c. Cu d. Ca

(A) a, c (B) a, d

(C) a, b (D) All of above

Q21. A, B, C and D are four metals. A can displace D from its salt solution, but B and C cannot displace D. B can displace hydrogen (H) for a dilute solution of a mineral acid, but C cannot. The reduction potentials of A, B, C, D and H (i.e., hydrogen) are in the order :

(A) C > H > B > D > A

(B) A > B > C > D > H

(C) A > D > B > H > C

(D) H > A > C > D > B

Balancing of Redox Reaction

Q22. In the balanced reaction;

$$xBrO_3^- + yCr^{+3} + zH_2O \longrightarrow aBr_2 + bHCrO_4^- + cH^+$$

the coefficients x, y and z are:-

(A) 6, 8, 22 (B) 6, 10, 22

(C) 6, 7, 14 (D) 7, 10, 20

Q23. For the redox reaction $MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$, the correct coefficients of the reactants for the balanced equation are :

	MnO_4^-	$C_2O_4^{2-}$	H^+
(A)	2	5	16
(B)	16	5	2
(C)	5	16	2
(D)	2	16	5

Q24. In the redox reaction :

$x\ KmnO_4 + y\ NH_3 \rightarrow KNO_3 + MnO_2 + KOH + H_2O$

(A) x = 4, y = 6 (B) x = 8, y = 6

(C) x = 3, y = 8 (D) x = 8, y = 3

Ion - electron Method of Balancing Redox Reactions

Q25. The reaction of $KMnO_4$ and HCl results in:-

(A) Oxidation of Mn and production of Cl_2

(B) Reduction of Mn and production of H_2

(C) Oxidation of Mn and production of H_2

(D) Reduction of Mn and production of Cl_2

Q26. In the balanced chemical reaction,

$IO_3^- + a\ I^- + bH^+ \rightarrow cH_2O + dI_2$

a, b, c and d respectively correspond to :-

(A) 5, 6, 3, 3 (B) 5, 3, 6, 3

(C) 3, 5, 3, 6 (D) 5, 6, 5, 5

Equivalent Weight

Q27. The number of Fe^{+2} ion oxidised by one mole of MnO_4^- ion is :-

(A) $\frac{1}{5}$ (B) $\frac{2}{3}$ (C) 5 (D) $\frac{3}{2}$

Q28. Number of moles of MnO_4^- required to oxidise one mole of ferrous oxalate completely in acid medium will be -

(A) 7.5 moles (B) 0.2 moles

(C) 0.6 moles (D) 0.4 moles

Q29. The equivalent mass of H_3PO_4 in the following reaction is :

$H_3PO_4 + Ca(OH)_2 \longrightarrow CaHPO_4 + 2H_2O$ [P = 31]

(A) 98 (B) 49

(C) 32.66 (D) 40

Q30. Equivalent weight of FeS_2 in the half reaction, $FeS_2 \rightarrow Fe_2O_3 + SO_2$ is :

(A) M/10 (B) M/11

(C) M/6 (D) M/1

Q31. The equivalent weight of an element is 4. Its chloride has a vapour density 59.25. The valency of the element is :-

(A) 4 (B) 3

(C) 2 (D) 1

Q32. Hydrazine reacts with KIO_3 in presence of HCl as;

$N_2H_4 + IO_3^- + 2H^+ + Cl^- \longrightarrow ICl + N_2 + 3H_2O$

The equivalent masses of N_2H_4 and KIO_3 respectively are :-

(A) 8, 87 (B) 8, 35.6

(C) 16, 53.5 (D) 8, 53.5

Q33. Equivalent weight of H_3PO_2 when it disproportionate in to PH_3 and H_3PO_3 is :-

(A) M (B) M/2

(C) M/4 (D) 3M/4

Q34. When potassium permagnate is titrated against ferrous ammonium sulphate in acidic medium the equivalent weight of potassium permagnate is :-

(A) $\frac{\text{Molecular weight}}{3}$

(B) $\frac{\text{Molecular weight}}{5}$

(C) $\frac{\text{Molecular weight}}{2}$

(D) $\frac{\text{Molecular weight}}{10}$

Q35. When CrI_3 oxidises to $Cr_2O_7^{2-}$ and IO_4^-, equivalent mass of CrI_3 will be :-

(A) $\frac{M}{33}$ (B) $\frac{M}{27}$ (C) $\frac{M}{28}$ (D) $\frac{M}{24}$

Q36. In the standarization of $Na_2S_2O_3$ using $K_2Cr_2O_7$ by iodometry the equivalent mass of $K_2Cr_2O_7$ is-

(A) $\frac{\text{M. mass}}{2}$ (B) $\frac{\text{M. mass}}{6}$

(C) $\frac{\text{M. mass}}{3}$ (D) Same as M. mass

Normality and Equiralent Concept

Q37. How many moles of hydrazine can be oxidized into N_2 using $\frac{2}{3}$ mole of bromate ions $(BrO_3^- \rightarrow Br^-)$:-

(A) $\frac{1}{3}$ (B) 1

(C) 1.5 (D) $\frac{2}{3}$

Q38. 6×10^{-3} mole $K_2Cr_2O_7$ reacts completely with 9×10^{-3} mole x^{n+} to give XO_3^- and Cr^{+3}. The value of n is :-

(A) 1 (B) 2

(C) 3 (D) 4

Q39. If 0.5 g of a metal on oxidation formed 0.79 g of its oxide, then equivalent weight of the metal is :-

(A) 10 (B) 14

(C) 18 (D) 22

Q40. Three solutions of HCl having normality 12 N, 6N and 2N are mixed to obtain a solution of 4N normality which among the following volume ratio is correct for above three components ?

(A) 1 : 2 : 6 (B) 2 : 1 : 9

(C) 1 : 2 : 4 (D) 1 : 1 : 5

Q41. How much of NaOH is required to neutralise 1500 cm^3 of 0.1 N HCl (At. wt. of Na =23).

(A) 4 g (B) 6 g

(C) 40 g (D) 60 g

Titration, Iodometry, Iodimetry

Q42. $KMnO_4$ reacts with ferrous ammonium sulphate according to the equation

$MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$, here 10 ml of 0.1 M $KMnO_4$ is equivalent to-

(A) 20 ml of 0.1 M $FeSO_4$

(B) 30 ml of 0.1 M $FeSO_4$

(C) 40 ml of 0.1 M $FeSO_4$

(D) 50 ml of 0.1 M $FeSO_4$

Q43. 100 ml of 0.1 N hypo decolourised iodine by the addition of χ g of crystalline copper sulphate to excess of KI. The value of 'χ' is (molecular wt. of $CuSO_4.5H_2O$ is 250).

(A) 5.0 g (B) 1.25 g

(C) 2.5 g (D) 4 g

Q44. 1.575 g of $(COOH)_2 . XH_2O$ are dissolved in water and the volume is made upto 250 ml. On titration 16.68 ml of this solution are required for the neutralisation of 25 ml of $\frac{N}{15}$ NaOH. Calculate the value of X :-

(A) 3 (B) 1

(C) 2 (D) 5

Q45. 5 ml of N HCl, 20 ml of N/2 H_2SO_4 and 30 ml of N/3 HNO_3 are mixed together and volume made to one litre. The normally of the resulting solution is :-

(A) $\frac{N}{5}$ (B) $\frac{N}{10}$ (C) $\frac{N}{20}$ (D) $\frac{N}{40}$

ACC

Q46. Fe shows an oxidation state of +1 in :-

(A) $Fe(CO)_5$

(B) $[Fe(H_2O)_5NO]SO_4$

(C) $Fe_4[Fe(CN)_6]_3$

(D) $FeCl_4^-$

Q47. Oxidation of Na is ?
(A) Addition of hydrogen to sodium
(B) Addition of oxygen to sodium
(C) Addition of chlorine to sodium
(D) All the above

Q48. A compound contains atoms A, B and C. The oxidation number of A is + 2, of B is + 5 and C is –2. the possible formula of the compound is-
(A) ABC_2 (B) $B_2(AC_3)_2$
(C) $A_3(BC_4)_2$ (D) $A_3(B_4C)_2$

Q49. Equivalent mass of oxidising agent in the reaction, $SO_2 + 2H_2S \rightarrow 3S + 2H_2O$ is -
(A) 32 (B) 64 (C) 16 (D) 8

Q50. 1 mole of equimolar mixture of ferric oxalate and ferrous oxalate will require x mole of $KMnO_4$ in acidic medium for complete oxidation, x is :-
(A) 0.5 mole (B) 0.9 mole
(C) 1.2 mole (D) 4.5 mole

Q51. Oxidising product of substance Na_3AsO_3 would be
(A) $As_2O_3^{3-}$ (B) AsO_3^{-3}
(C) AsO_2^{-4} (D) AsO_4^{-3}

Q52. Bleaching action of SO_2 is due to :
(A) Reduction
(B) Oxidation
(C) Hydrolysis
(D) Acidic nature

Q53. In alkaline medium ClO_2 oxidises H_2O_2 to O_2 and itself gets reduced to Cl^-. How many moles of H_2O_2 are oxidised by 1 mole of ClO_2 ?
(A) 1 (B) 1.5
(C) 2.5 (D) 3.5

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which molecule have underlined atoms in their highest oxidation state?
(A) $H_2\underline{S_2}O_8$ (B) $\underline{P_4}O_{10}$
(C) $\underline{S}O_2$ (D) $\underline{Mn_2}O_7$

Q2. In which of the following compounds further oxidation is possible?
(A) FeO (B) Fe_2O_3
(C) SO_2 (D) CrO_3

Q3. Which of the following compounds is/are oxidising agent?
(A) $KMnO_4$ (B) KI
(C) $K_2Cr_2O_7$ (D) $K_2C_2O_4$

Q4. In which of the following species the O.S. of the carbon is/are equal to + 2 only
(A) $H - C \equiv N$ (B) $H - N \rightleftarrows C$
(C) $C \rightleftarrows O$ (D) $O=C=C=C=O$

Q5. Which of the following are (is) example of disproportionation reaction ?
(A) $HgO \longrightarrow Hg + O_2$
(B) $KClO_3 \longrightarrow KCl + O_2$
(C) $KClO_3 \longrightarrow KClO_4 + KCl$
(D) $Cl_2 + OH^- \longrightarrow ClO^- + Cl^- + H_2O$

Q6. Which of the following is (are) not the redox reaction (s) ?

(A) $NaBr + HCl \longrightarrow NaCl + HBr$

(B) $HBr + AgNO_3 \longrightarrow AgBr + HNO_3$

(C) $H_2 + Br_2 \longrightarrow 2HBr$

(D) $Na_2O + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O$

Q7. Which of the following statements is/are correct?

(A) molar ratio $= \dfrac{1}{\text{n} - \text{factor ratio}}$

(B) no. of equivalents = moles × n-factor

(C) n-factor of H_3PO_3 is 3

(D) n-factor of H_3BO_3 is 3

Q8. For the reaction :

$H_3PO_3 + Ba(OH)_2 \longrightarrow BaHPO_3 + 2H_2O$

The correct statement (s) is/are

(A) The equivalent mass of H_3PO_3 is 41

(B) $BaHPO_3$ is the acid salt

(C) $BaHPO_3$ is normal salt

(D) 1 mole of H_3PO_3 is completely neutralised by 1.5 moles of $Ba(OH)_2$

Q9. Choose the correct statements –

(A) Copper sulphate solution can be stored in a zinc vessel.

(B) Copper sulphate solution can be stored in a nickel vessel.

(C) Copper sulphate solution can be stored in a silver vessel.

(D) Copper sulphate solution can be stored in a gold vessel.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

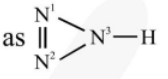
Paragraph For Q. 10 to 14

Oxidation reduction reaction are those in which oxidation reduction take place simultaneously. Oxidation and reduction are determined by the oxidation number. The oxidation state of any element can never be fractional, if oxidation number of any element comes out be in fraction, it is the average oxidation number of that element which is present in different oxidation states.

Q10. The Fe_3O_4 is the mixed oxides containing FeO & Fe_2O_3, therefore oxidation states of Fe in Fe_3O_4 are-

(A) +2 and +3 (B) +8/3

(C) +2 (D) +3

Q11. The structural formula of hydrazoic acid is represented as 

Therefore oxidation states of N^1, N^2 and N^3 respectively are -

(A) 0, 0, 3 (B) 0, 0, – 1

(C) 1, 1, – 3 (D) – 3, – 3, – 3

Q12. The equivalent weight of $Na_2S_2O_3$ in the following reaction is

$Na_2S_2O_3 + 2HCl \rightarrow 2NaCl + H_2O + S + SO_2$

Given : molar mass of $Na_2S_2O_3$ = M

(A) M (B) $\dfrac{M}{2}$ (C) $\dfrac{M}{3}$ (D) $\dfrac{3M}{4}$

Q13. Oxidation state of Cl in $CaOCl_2$ is?

(A) 0 (B) –1

(C) +1 and –1 (D) +1

Q14. In which of the following reactions H_2O_2 acts as a reducing agent.

(A) $H_2O_2 + 2H^+ + 2e^- \longrightarrow 2H_2O$

(B) $H_2O_2 - 2e^- \longrightarrow O_2 + 2H^+$

(C) $H_2O_2 + 2e^- \longrightarrow 2OH^-$

(D) $H_2O_2 + 2OH^- - 2e^- \longrightarrow O_2 + 2H_2O$

Paragraph for Q. 15 to 17

Observe the following reactions and answer the following questions.

(i) $Zn(s) + PbCl_2(aq) \longrightarrow Pb(s) + ZnCl_2(aq)$

(ii) $2Fe^{3+}(aq) + 2I^-(aq) \longrightarrow I_2(aq) + 2Fe^{2+}(aq)$

(iii) $2Na(s) + Cl_2(g) \longrightarrow 2NaCl(s)$

(iv) $Mg(s) + Cl_2(g) \longrightarrow MgCl_2(s)$

(v) $Zn(s) + 2H^+(aq) \longrightarrow Zn^{2+}(aq) + H_2(g)$

Q15. Which reactants are oxidising agents –

(1) Pb^{2+} (2) Fe^{+3} (aq)

(3) $Cl_2(g)$ (4) H^+ (aq)

(A) 1, 2, 3, 4 (B) 1, 2, 3

(C) 1, 3, 4 (D) 1, 2

Q16. Which reactants are reducing agents –

(1) Zn (2) I^- (aq)

(3) $Cl_2(g)$ (4) H^+ (aq)

(A) 1, 2, 3, 4 (B) 1, 2, 3

(C) 1, 3, 4 (D) 1, 2

Q17. Which of the following species can be oxidised –

(1) Na (2) Mg

(3) I^- (4) H^+

(A) 1, 2, 3, 4 (B) 1, 2, 3

(C) 1, 3, 4 (D) 1, 2

MATCHING LIST TYPE

Each question has TWO (02) ***matching lists: LIST I and LIST II****. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q18. Compounds O.N.

List-I	**List-II**
(A) $K\overset{*}{M}nO_4$	(1) +4
(B) $\overset{*}{N}i(CO)_4$	(2) +7
(C) $[\overset{*}{P}t(NH_3)Cl_2]Cl_2$	(3) 0
(D) $Na_2\overset{*}{O}_2$	(4) –1

The correct code for the O.N. of asterisked atom would be :

	A	B	C	D
(A)	1	2	3	4
(B)	4	3	2	1
(C)	2	3	1	4
(D)	4	1	2	3

Q19. Match the Column-I with Column-II –

Column-I (Redox process)	**Column-II (n-factor for underlined species)**
(a) $\underline{As_2S_3} \rightarrow AsO_3^- + SO_4^{2-}$	(p) 28
(b) $\underline{I_2} \rightarrow I^- + IO_3^-$	(q) 4/3
(c) $\underline{H_3PO_2} \rightarrow PH_3 + 2H_3PO_3$	(r) 1
(d) $\underline{H_3PO_2} + NaOH \rightarrow NaH_2PO_2 + H_2O$	(s) 5/3

(A) a-s; b-p; c-q; d-r
(B) a-p; b-r; c-q; d-s
(C) a-s; b-p; c-r; d-q
(D) a-p; b-s; c-q; d-r

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q20. Match the half-reactions (in column- I) with change in oxidation number (in column- II) :

Column I	**Column II**
(A) $Cl^- \longrightarrow ClO_4^-$	(P) 2
(B) $Cr^{3+} \longrightarrow CrO_5$	(Q) 8
(C) $H_2O_2 \longrightarrow O_2$	(R) 0
(D) $CrO_2^{2+} \longrightarrow CrO_4^{2-}$	(S) 3

Q21. Match the column correctly –

Column I	**Column II**
(A) Change of $Cr_2O_7^{2-}$ to CrO_4^{-2}	(P) Oxidation
(B) Change of BrO_3^{-1} to BrO_4^-	(Q) Reduction
(C) Change of $S_4O_6^{-2}$ to $S_2O_3^{-2}$	(R) Neither reduction nor oxidation
(D) Change of Cu_2O to CuO	(S) loss/gain of $2e^-$

Q22. Match the column correctly –

Column I	**Column II**
(A) $O_2^{-2} \longrightarrow O_2 + O^{2-}$	(P) redox reaction
(B) $CrO_4^{2-} + H^+ \longrightarrow$	(Q) One of the species undergoes only oxidation
(C) $MnO_4^- + NO_2^- + H^+ \longrightarrow$	(R) One of the species undergoes only reduction
(D) $NO_3^- + H_2SO_4 + Fe^{2+} \longrightarrow$	(S) disproportionation.

Q23.

Column I	Column II
(A) $(NH_4)_2Cr_2O_7 \rightarrow N_2 + Cr_2O_3 + 4H_2O$	(P) Intermolecular redox reaction
(B) $PbO_2 + H_2O \rightarrow PbO + H_2O_2$	(Q) Disproportionation
(C) $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$	(R) Intramolecular redox reaction
(D) $Cl_2 + 2OH^- \rightarrow ClO^- + Cl^- + H_2O$	(S) Metal displacement

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q24. How many electrons are involved in following reaction :

$$Cr_2O_7^{2-} + Fe^{+2} + C_2O_4^{-2} \rightarrow 2Cr^{+3} + Cr^{+3} + CO_2$$

Q25. The value of n in the equation

$$Cr_2O_7^{-2} + 14H^+ + ne^- \rightarrow 2Cr^{+3} + 7H_2O$$

is ?

Q26. Calculate equivalent weight of S in SO_3^{-2} ?

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q27. **Statement 1 :**

Fluorine exists only in –1 oxidation state.

Statement 2 :

Fluorine has $2s^2\ 2p^5$ configuration of valence shell.

SUBJECTIVE TYPE

Q28. Select the species acting as reductant and oxidant in the reaction given below :

(i) $PCl_3 + Cl_2 \longrightarrow PCl_5$

(ii) $AlCl_3 + 3K \longrightarrow Al + 3KCl$

(iii) $SO_2 + 2H_2S \longrightarrow 3S + H_2O$

(iv) $BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2NaCl$

(v) $3I_2 + 6NaOH \longrightarrow NaIO_3 + 5NaI + 3H_2O$

Q29. Write complete balanced equation for the following in basic medium by ion-electron method:

(a) $H_2 + ReO_4^- \longrightarrow ReO_2 + H_2O$

(b) $Cu^{+2} + I^- \longrightarrow Cu^+ + I_2$

(c) $KOH + K_4Fe(CN)_6 + Ce(NO_3)_4 \longrightarrow Fe(OH)_3 + Ce(OH)_3 + K_2CO_3 + KNO_3 + H_2O$

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. In S_8, $S_2O_7^{2-}$ and S_2Cl_2 the oxidation state of sulphur is respectively :-

(A) +8, + 6, + 1 (B) 0, +6, +2

(C) 0, +8, +1 (D) 0, +6, +1

Q2. In the ethylene molecule, the two carbon atoms have the oxidation numbers :-

(A) – 1, – 1 (B) – 2, – 2

(C) – 1, – 2 (D) + 2, – 2

Q3. Calculate the correct increasing order of average oxidation number of sulphur in : S_8,$S_2O_3^{2-}$, $S_2O_7^{-2}$, $S_4O_6^{-2}$, SO_3^{-2} :-

(A) $S_8 < S_2O_3^{-2} < S_4O_6^{-2} < SO_3^{-2} < S_2O_7^{-2}$

(B) $S_2O_3^{-2} < S_4O_6^{-2} < S_8 < SO_4^{-2} < S_2O_7^{-2}$

(C) $S_4O_6^{-2} < S_2O_3^{-2} < S_8 < SO_4^{-2} < S_2O_7^{-2}$

(D) $S_8 < S_4O_6^{-2} < S_2O_3^{-2} < SO_3^{-2} < S_2O_7^{-2}$

Q4. A compound contain atoms X, Y, Z. The oxidation number of X is +2, Y is +5 and Z is –2. The possible formula of the compound is :-

(A) XY_1Z_2 (B) $Y_2(XZ_3)_2$

(C) $X_3(YZ_4)_2$ (D) $X_3(Y_4Z)_2$

Q5. The oxidation number of nickel in $K_4Ni(CN)_6$ is:-

(A) + 1 (B) + 2

(C) – 1 (D) 0

Q6. Which of the following has been arranged in order of increasing oxidation number of nitrogen :

(A) $NH_3 < N_2O_5 < NO < N_2$

(B) $NO_2^+ < NO_3^- < NO_2^- < N_3^-$

(C) $NH_4^+ < N_2H_4 < NH_2OH < N_2O$

(D) $NO_2 < NaN_3 < NH_4^+ < N_2O$

Q7. The oxidation number of chromium in blue per chromate (CrO_5) :-

(A) + 3 (B) + 5

(C) + 6 (D) 4

Q8. 2 mole of N_2H_4 loss 16 mole of electron is being converted to a new compound X. Assuming that all of the N appears in the new compound. What is the oxidation state of 'N' in X :-

(A) – 1 (B) – 2

(C) + 2 (D) + 4

Q9. The resonating structures of cyanate ion are

$O = C = N^- \longleftrightarrow {}^-O{-}C \equiv N \longleftrightarrow {}^+O \equiv C{-}N^-$.

(i) (ii) (iii)

The correct set of oxidisation states of O, C & N respectively in the most stable structure out of the above is :

(A) (i), –2, +4, –3 (B) (ii), –2, + 4, –3

(C) (iii), –2, +4, –3 (D) (ii), 0, +4, –5

Q10. The oxidation number of C in CH_4, CH_3Cl, CH_2Cl_2, $CHCl_3$ and CCl_4 are respectively :

(A) +4, +2, 0, –2, –4

(B) +2, +4, 0, –4, –2

(C) –4, –2, 0, +2, +4

(D) –2, –4, 0, +4,. +2

Q11. Oxidation number of carbon in carbon suboxide (C_3O_2) is :

(A) $\frac{+2}{3}$ (B) $\frac{+4}{3}$

(C) +4 (D) $\frac{-4}{3}$

Q12. In the brown ring complex $[Fe(H_2O)_5NO^+]SO_4$,
the oxidation number of Fe is :
(A) +1 (B) +2
(C) +3 (D) +4

Q13. Which is correct statement regarding CrO_5
A : Oxidation number of Cr is +6.
B : Four O-atoms of molecule are involved in peroxide linkage.
C : Only one O^- atom has O.N. = –2
(A) All are correct
(B) B, C are correct
(C) A, C are correct
(D) Only B is correct

Q14. The oxidation state of the most electronegative element in the products of the reaction between BaO_2 and H_2SO_4 are :
(A) 0 and 1
(B) –1 and –2
(C) –2 and 0
(D) –2 and +1

Q15. Oxidation state of nitrogen is incorrectly given for -

Compound	Oxidation state
(i) $[Co(NH_3)_5Cl]Cl_2$	0
(ii) NH_2OH	–1
(iii) $(N_2H_5)_2SO_4$	–2
(iv) Mg_3N_2	–3

(A) i (B) ii
(C) iii (D) iv

Q16. When KMnO4 acts as an oxidising agent and ultimately forms MnO_4^{2-}, MnO_2, Mn_2O_3 and Mn^{2+}, then the number of electrons transferred in each case respectively is :
(A) 4, 3, 1, 5 (B) 1, 5, 3, 7
(C) 1, 3, 4, 5 (D) 3, 5, 7, 1

Q17. The reaction of $KMnO_4$ and HCl results in :
(A) Oxidatin of Mn in $KMnO_4$ and production of Cl_2
(B) Reduction of Mn in $KMnO_4$ and production of H_2
(C) Oxidation of Mn in $KMnO_4$ and production of H_2
(D) Reduction of Mn in $KMnO_4$ and production of Cl_2

Q18. In the reaction $H_2S + I_2 \rightarrow 2\,HI + S$
(A) H_2S is reduced
(B) H_2S is oxidised
(C) I_2 is oxidised
(D) I_2 is acting as reducing agent

Q19. One mole of iron [55.8 gm], when oxidised to +2 oxidation state gives up :
(A) $1N_A$ electron
(B) $2N_A$ electron
(C) $3N_A$ electron
(D) 0.5 mole of electron

Q20. In which of the following reaction H_2O_2 does not act. as an oxidising agent-
(A) $2FeSO_4 + H_2SO_4 + H_2O_2 \rightarrow Fe_2(SO_4)_3 + 2H_2O$
(B) $2Cr(OH)_3 + 4NaOH + 3H_2O_2 \rightarrow 2Na_2CrO_4 + 8H_2O$
(C) $2K_3[Fe(CN)_6] + 2KOH + H_2O_2 \rightarrow 2K_4[Fe(CN)_6] + 2H_2O + O_2$
(D) $2KI + H_2O_2 \rightarrow 2KOH + I_2$

Q21. The number of electrons required to balance the following equation,
$NO_3^- + 4H^+ + e^- \longrightarrow 2H_2O + NO$ is -
(A) 5 (B) 4
(C) 3 (D) 2

Q22. What is the value of *n* in the following equation
$Cr(OH)_3 + OH^- \rightarrow CrO_4^{2-} + H_2O + ne$?
(A) 3 (B) 6 (C) 5 (D) 2

Q23. In the reaction, $xVO + yFe_2O_3 \rightarrow FeO + V_2O_5$. What is the value of x and y respectively :-

(A) 1, 1 (B) 2, 3

(C) 3, 2 (D) 4, 2

Q24. In the reaction, $xBrO_3^- + yCr^{3+} + zH_2O \rightarrow Br_2 + HCrO_4^- + H^+$, the coefficients x, y and z are:

(A) 6, 8, 22 (B) 6, 10, 22

(C) 6, 10, 11 (D) 6, 10, 20

Q25. The balanced form for an equation $K_2Cr_2O_7 + HCl \rightarrow KCl + CrCl_3 + H_2O + Cl_2$ will involve :-

(A) 14 HCl and 7 H_2O

(B) 2 KCl and 2 $CrCl_3$

(C) $K_2Cr_2O_7$ and 3 Cl_2

(D) All of these

Q26. Number of moles of $KMnO_4$ required to oxidize one mole of $Fe_2(C_2O_4)_3$ in acidic medium is-

(A) 1.2 (B) 0.167

(C) 0.2 (D) 0.4

Q27. $KMnO_4 + HCl \rightarrow MnCl_2 + Cl_2 + KCl + H_2O$

In the above reaction how many moles of H_2O would be formed for each mole Cl_2 liberated.

(A) 5/4 (B) 4/5

(C) 5/8 (D) 8/5

Q28. How many electrons should X_2H_4 liberate so that in the new compound X shows oxidation number of $-\frac{1}{2}$ [E.N. X > H]

(A) 2 (B) 3

(C) 4 (D) 6

Q29. Dichromate ion in acidic medium oxidizes stannous ion as :

$xSn^{2+} + yCr_2O_7^{-2} + zH^+ \Rightarrow aSn^{+4} + bCr^{+3} + cH_2O$

(I) The value of x : y is 1 : 3

(II) The value of x + y + z may 18

(III) a : b is 3 : 2

(IV) The value of z-c is 7

The correct code is -

(A) I, III, IV (B) II, III, IV

(C) I, II, IV (D) I, III

Q30. Equivalent weight of H_3PO_3 when it dispropertinate into PH_3 and H_3PO_4 is-:-

(A) M (B) M/2

(C) 3M/4 (D) 2M/3

Q31. In the reaction, $2S_2O_3^{2-} + I_2 \rightarrow S_4O_6^{2-} + 2I^-$, the eq. wt. of $Na_2S_2O_3$ is equal to its :

(A) Mol. wt. (B) Mol. wt./2

(C) 2 x Mol. wt. (D) Mol. wt./6

Q32. What is the equivalent weight of $C_{12}H_{22}O_{11}$ in the following reaction ?

$C_{12}H_{22}O_{11} + 36HNO_3 \longrightarrow 6H_2C_2O_4 + 36NO_2 + 23\ H_2O$

(A) $\frac{342}{36}$ (B) $\frac{342}{12}$

(C) $\frac{342}{22}$ (D) $\frac{342}{3}$

Q33. What would be the equivalent weight of the reductant in the reaction

$2[Fe(CN)_6]^{3-} + H_2O_2 + OH^- \rightarrow 2[Fe(CN)_6]^{4-} + 2H_2O + O_2$

(A) 17 (B) 212

(C) 34 (D) 32

Q34. w_1g of metal combines with w_2g of oxygen forming it's oxide. The equivalent weight of metal is :-

(A) $\frac{8w_1}{w_2}$ (B) $\frac{8w_2}{w_1}$

(C) $\frac{8w_1}{w_2 - w_1}$ (D) $\frac{8w_2}{w_2 - w_1}$

Q35. Equivalent weight of $KMnO_4$ in acidic medium and neutral (M= molecular wt.) medium respectively-

(A) $\frac{M}{3}, \frac{M}{5}$ (B) $\frac{M}{3}, \frac{M}{3}$

(C) $\frac{M}{5}, \frac{M}{3}$ (D) $\frac{M}{4}, \frac{M}{5}$

Q36. 1.25 g of a solid dibasic acid is completely neutralised by 25 mL of 0.25 molar $Ba(OH)_2$ solution. Molecular mass of the acid is :-

(A) 100 (B) 150

(C) 120 (D) 200

Q37. The equiv. wt. of hypo in the reaction

$Na_2S_2O_3 + Cl_2 + H_2O \longrightarrow Na_2SO_4 + H_2SO_4 + HCl$ is-

(A) $\frac{\text{Mol. wt.}}{2}$ (B) $\frac{\text{Mol. wt.}}{4}$

(C) $\frac{\text{Mol. wt.}}{1}$ (D) $\frac{\text{Mol. wt.}}{8}$

Q38. The equivalent wt. of the salt $KHC_2O_4 . H_2C_2O_4 . 4H_2O$ (to be used as a reducing agent) is-

(A) $\frac{\text{Mol. wt.}}{1}$ (B) $\frac{\text{Mol. wt.}}{2}$

(C) $\frac{\text{Mol. wt.}}{3}$ (D) $\frac{\text{Mol. wt.}}{4}$

Q39. How many moles of stannous oxalate can be oxidised into Sn^{+4} and CO_2 using 1 mole of $K_2Cr_2O_7$:

(A) 0.5 (B) 1

(C) 1.5 (D) 2

Q40. Calculate the number of milli moles of SO_2. If in the following reaction 10 ml of 0.1 M $KMnO_4$ solution are required for titration.

$SO_2 + MnO^-_4 \rightarrow SO_4^{-2} + Mn^{+2}$

(A) 2.5 (B) 0.5

(C) 1.25 (D) None of these

Q41. A solution of $Na_2S_2O_3$ is standardized iodometrically against 0.167 g of $KBrO_3$ where BrO_3^- changes to Br^-. This process requires 45 mL of the $Na_2S_2O_3$ solution. What is the strength of the $Na_2S_2O_3$?

[Mw. of $KBrO_3$ = 167]

(A) $\frac{2}{15}$N (B) $\frac{2}{30}$N

(C) $\frac{1}{30}$N (D) $\frac{1}{60}$N

Q42. The difference in the oxidation numbers of two types of carbon atom in C_3O_2 is :-

(A) 4 (B) 5

(C) 2 (D) 1

Q43. In the reaction $P + NaOH \rightarrow PH_3 + NaH_2PO_2$

(A) P is oxidised only

(B) P is reduced only

(C) P is oxidized as well as reduced

(D) Na is reduced

Q44. Sulphur forms the chlorides S_2Cl_2 and SCl_2. If the equivalent weight of sulphur in SCl_2 is 16, then the equivalent weight of sulphur is S_2Cl_2 is :-

(A) 8 (B) 16

(C) 32 (D) 64

Q45. Ratio of moles of Fe(II) oxidised by equal volumes of equimolar $KMnO_4$ and $K_2Cr_2O_7$ solutions in acidic medium will be :-

(A) 5 : 3 (B) 1 : 1

(C) 1 : 2 (D) 5 : 6

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following can be used both as reductant and oxidant?

(A) HNO_2 (B) SO_2

(C) O_2 (D) CO

Q2. No reaction occurs in which of following equations

(A) $I^- + Fe^{+2} \longrightarrow$

(B) $F_2 + 2NaCl \longrightarrow$

(C) $Cl_2 + 2NaF \longrightarrow$

(D) $I_2 + 2NaBr \longrightarrow$

Q3. Identify the disproportionation reaction (s)-

(A) $KIO_3 + 8\ KI + 3H_2SO_4 \longrightarrow 3KI_3 + 3K_2SO_4 + 3H_2O$

(B) $Cu_2O + 2H^+ \longrightarrow Cu + Cu^{2+} + H_2O$

(C) $2H_2O_2 \longrightarrow 2H_2O + O_2$

(D) $3MnO_4^{-2} + 2H_2O \longrightarrow MnO_2 + 2MnO_4^- + 4OH^-$

Q4. Excess of KI reacts with $CuSO_4$ solution and then $Na_2S_2O_3$ solution is added to it. Which of the statements is/are correct for this reaction ?

(A) Cu_2I_2 is formed

(B) Evolved I_2 is reduced

(C) $Na_2S_2O_3$ is oxidised

(D) CuI_2 is formed

Q5. 20 volumes of H_2O_2 is equal to -

(A) 20 % H_2O_2 by mass

(B) 6 % H_2O_2 by mass

(C) 1.764 N

(D) 3.528 N

Q6. Which of the following compounds have n-factor equals to 2?

(A) H_2SO_4 (B) H_3PO_4

(C) H_3PO_3 (D) N_2H_4

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 7 to 8

The concentration of solution can be expressed in number of ways such that normality, molarity, mole friction, strength % by weight, % by volume and % by strength. the molarity of Ionic compound is usually expressed as formality because we use formula weight of ionic compound.

Q7. Number of oxalite ions in 100 ml of 0.1N oxalic acid is :-

(A) $\frac{N_A}{100}$ (B) $\frac{N_A}{20}$

(C) $\frac{N_A}{200}$ (D) $\frac{N_A}{1000}$

Q8. The normality of 0.3 M H_3BO_3 is :-

(A) 0.3 N (B) 0.15 N

(C) 0.6 N (D) 0.9 N

Paragraph For Q. 9 to 11

$KMnO_4$ solution is to be standardized by titration against $As_2O_3(s)$. A 0.1097 g sample of As_2O_3 requires 26.10 ml of the $KMnO_4$ solution for its titration. The concentration of Mn^{2+} can be determined by titration with $KMnO_4$ solution in neutral medium resulting in formation of MnO_2. A 15.00 ml solution of Mn^{2+} requires 20.06 ml of the $KMnO_4$ solution. (As = 75, Mn = 55)

Q9. What is the normality of the $KMnO_4$ solution?

(A) 0.085 (B) 0.85

(C) 1.085 (D) 5.032

Q10. What is the molarity of the $KMnO_4$ solution?

(A) 0.017M (B) 0.17 M

(C) 1.7 M (D) 2.34 M

Q11. Calculate the concentration of Mn^{2+}.

(A) 0.0341 M (B) 0.341 M

(C) 1.231 M (D) 5.124 M

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q12.

Column- I	**Column-II**
(A) 2 mole electrons transfer	(P) $C_2O_4^{2-} \rightarrow CO_2$
(B) 3 mole of electrons transfer	(Q) $Fe_3O_4 \rightarrow Fe^{2+}$
(C) 4 mole of electrons transfer	(R) $CrO_4^{2-} \rightarrow Cr^{3+}$
(D) Reduction reactions	(S) $MnO_4^- \rightarrow MnF_4^-$

Q13. Match the following :

Column I	**Column II**
(A) Change of NO to NO_3^-	(P) gain of $2e^-$
(B) Change of P_4 to $H_2PO_2^-$	(Q) gain of $4e^-$
(C) Change of ClO^- to Cl^-	(R) Loss of $4e^-$
(D) Change of $S_2O_3^{2-}$ to S	(S) loss of $3e^-$

Q14.

Column-I	**Column-II**
(A) $SCN^- \rightarrow SO_4^{-2} + HCN$	(P) Disproportionation reaction
(B) $KClO_3 \rightarrow KClO_4 + KCl$	(Q) n-factor = 6
(C) $K_2Cr_2O_7 \xrightarrow{H^+} Cr_2(SO_4)_3 + Cr_2O_3$	(R) Equivalent mass = $\frac{\text{molecular mass}}{11}$
(D) $FeS_2 \longrightarrow Fe^{3+} + SO_2$	(S) Two elements show change in oxidation state

TRIPLE MATCH TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Q15.

Column I Compound	**Column II Oxidation No. underlined atom**	**Column-III**
(A) $K_2\underline{C_2}O_4$	(1) + 7	(P) Mohr's salt
(B) $K\underline{Mn}O_4$	(2) + 2	(Q) n-factor = 2
(C) $\underline{Fe}(SO_4)(NH_4)_2 3O_4.6H_2O$	(3) + 6	(R) underlined atom in heighest oxidation state
(D) $K_2\underline{Cr_2}O_7$	(4) + 3	(S) organe colour solution

Codes :

	A	B	C	D
(A)	4, P	1, R	3, S	2, Q
(B)	4, Q	1, R	2, P	3, S
(C)	3, Q	2, S	4, P	1, R
(D)	2, S	1, P	3, Q	4, R

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q16. Find oxidation number of Fe in FeS_2 -

Q17. The oxidation number of Mn in the product of alkaline oxidative fusion of MnO_2 is -

Q18. The normality of 2M phosphorus acid (H_3PO_3) is -

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q19. **Statement 1 :** Li is the strongest reducing agent, and Li^+ is the weakest oxidising agent.

Statement 2 : F_2 is the strongest oxidising agent, and F^- is the weakest reducing agent.

SUBJECTIVE TYPE

Q20. Complete the following redox equations and balance them.:

(a) $KnO_2 + Kl + HCl \longrightarrow$ + --------+ -------+ H_2O+ NO

(b) $Na_2S + HCl + SO_2 \longrightarrow$ + --------+ -------

(c) $As_2S_3 + NHO_3$ (Conc)+ $H_2O \longrightarrow$ + --------+-------+ NO

(d) $[Fe(CN)_6]^{3-} + H_2O_2 + OH^- \longrightarrow$ + --------+-------

Q21. Write complete balanced equation for the following in acidic medium by ion-electron method:

(a) $S_2O_3^{-2} + Sb_2O_5 \longrightarrow SbO + H_2SO_3$

(b) $IO_4^- + I^- + H^+ \longrightarrow I_2 + H_2O$

(c) $MnO_4^{-2} \longrightarrow MnO_4^{-1} + MnO_2$

Q22. Balance the following redox reactions by ion – electron method :

(a) $MnO_4^-(aq) + I^-(aq) \rightarrow MnO_2(s) + I_2(s)$ (in basic medium)

(b) $MnO_4^-(aq) + SO_2(g) \rightarrow Mn^{2+}(aq) + HSO_4^-(aq)$ (in acidic solution)

(c) $H_2O_2(aq) + Fe^{2+}(aq) \rightarrow Fe^{3+}(aq) + H_2O(l)$ (in acidic solution)

(d) $Cr_2O_7^{2-} + SO_2(g) \rightarrow Cr^{3+}(aq) + SO_4^{2-}(aq)$ (in acidic solution)

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
C	C	A	D	C	D	A	A	B	D
11	12	13	14	15	16	17	18	19	20
B	D	C	C	C	B	D	B	B	A
21	22	23	24	25	26	27	28	29	30
A	B	A	D	D	A	C	C	B	B
31	32	33	34	35	36	37	38	39	40
B	D	D	B	B	B	B	A	B	A
41	42	43	44	45	46	47	48	49	50
B	D	C	C	D	B	D	C	C	B
51	52	53							
D	A	C							

Ex-1A

1. A, B, D **2.** A, C **3.** A, C **4.** A, B, C **5.** C, D

6. A, B, D **7.** B **8.** A, C **9.** C, D **10.** A

11. B **12.** B **13.** C **14.** B, D **15.** A

16. D **17.** B **18.** C **19.** D

20. (A) → Q; (B) → S; (C) → P; (D) → R

21. (A) → R; (B) → P, S; (C) → Q, S; (D) → P, S

22. (A) → P, S ; (B) → R ; (C) → P, Q, R ; (D) → P, Q, R

23. (A) → R ; (B) → P ; (C) → P, S ; (D) → Q

24. 6 **25.** 6 **26.** 8 **27.** B

Ex-2

1	2	3	4	5	6	7	8	9	10
D	B	A	C	B	C	C	C	B	C
11	12	13	14	15	16	17	18	19	20
B	A	A	B	A	C	D	B	B	C
21	22	23	24	25	26	27	28	29	30
C	B	B	B	D	D	D	B	B	C
31	32	33	34	35	36	37	38	39	40
A	A	A	A	C	D	D	D	C	A
41	42	43	44	45					
A	C	C	C	D					

Ex-2A

1. A, B, C, D **2.** A, C, D **3.** B, C, D **4.** A, B, C **5.** B, D

6. A, C, D **7.** C **8.** A **9.** A **10.** A

11. A

12. (A) → P,Q; (B) → R; (C) → S; (D) → Q, R,S

13. (A) → S ; (B) → R ; (C) → P ; (D) → Q

14. (A) → Q, S ; (B) → P ; (C) → Q ; (D) → R, S

15. B **16.** 2 **17.** 6 **18.** 4N **19.** B